

SPECIAL TOPIC · the rarest words

Hapax Legomena — concepts the Qur'an says exactly ONCE

Two kinds of 'once', computed from Book6

A hapax legomenon appears a single time in a corpus. The Qur'an has two levels: a once-only ROOT (the whole consonantal family occurs once) and a once-only FORM (a specific surface word occurs once, though its root may be common). 408 roots and 3,027 forms are hapax.

Every example carries its address; we report the count and the address, and decline to over-read.

Two levels of 'once-only'

the corpus, by how rare its units are

ROOT hapax — 408

the whole 3-letter family occurs once e.g. (112:2) صمد

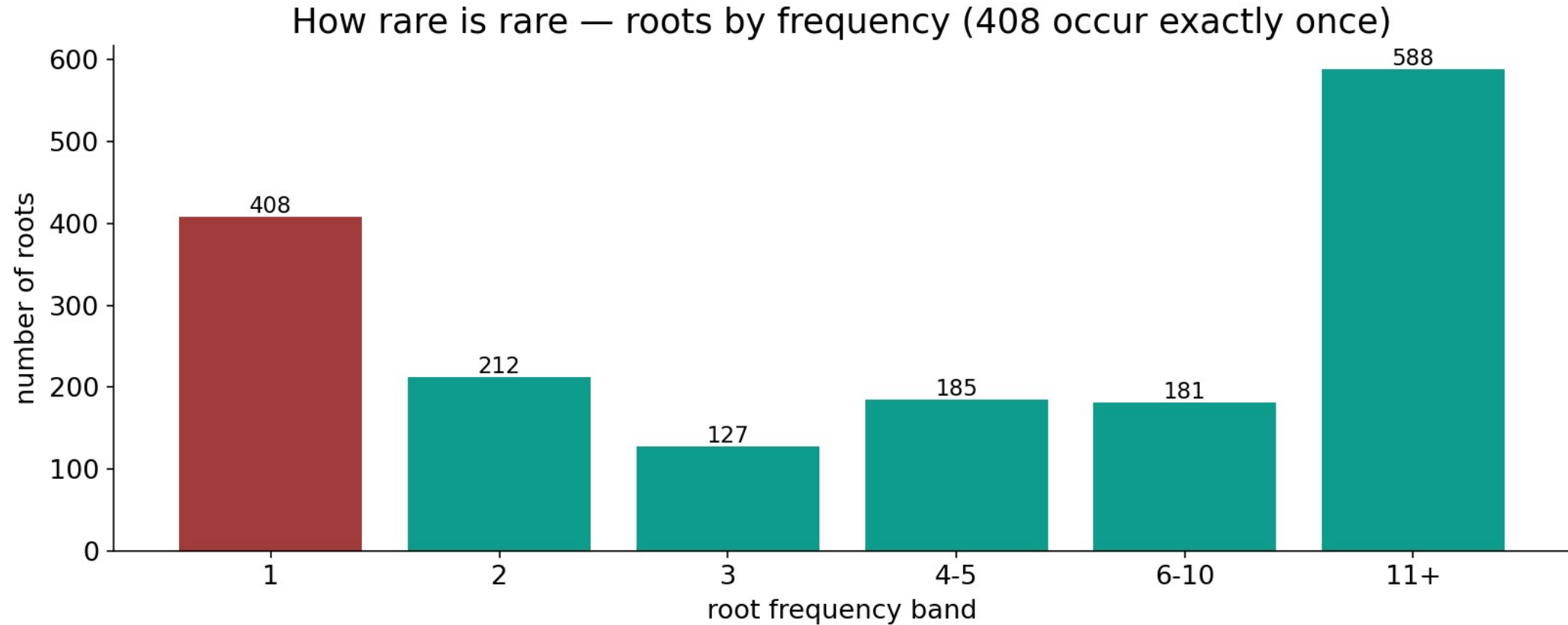
FORM hapax — 3,027

one surface word occurs once; root may be common e.g. كوثر (108:1, root كثر 167×)

Why distinguish them

A once-only ROOT is a concept the Qur'an names a single time. A once-only FORM is a unique wording of a possibly-common idea. The two answer different questions — unique concepts vs unique expressions — so we count and present them separately, both from Book6.

How rare is 'rare'? — roots by frequency



In the data — of 1,701 roots: 408 occur once, 212 twice, 127 thrice; only 588 occur 11+ times. Rarity is the NORM, not the exception — the hapax sit at the tip of a steep long tail.

Two flagship examples

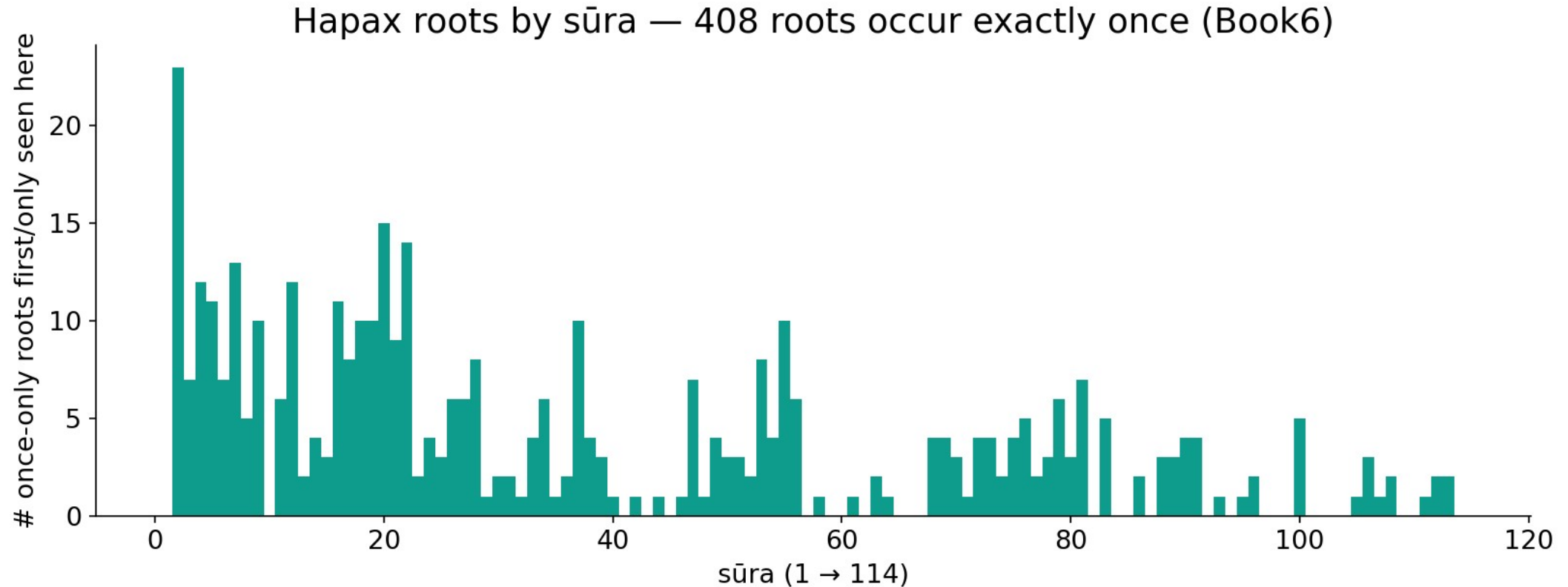
صمد — a once-only ROOT (112:2)

ص-م-د occurs once, in al-Ikhlāṣ: 'Allāh, aṣ-Ṣamad' — the Eternal, Absolute, self-sufficient. A unique root for a unique description of God. (root freq = 1.)

كوثر — a once-only FORM (108:1)

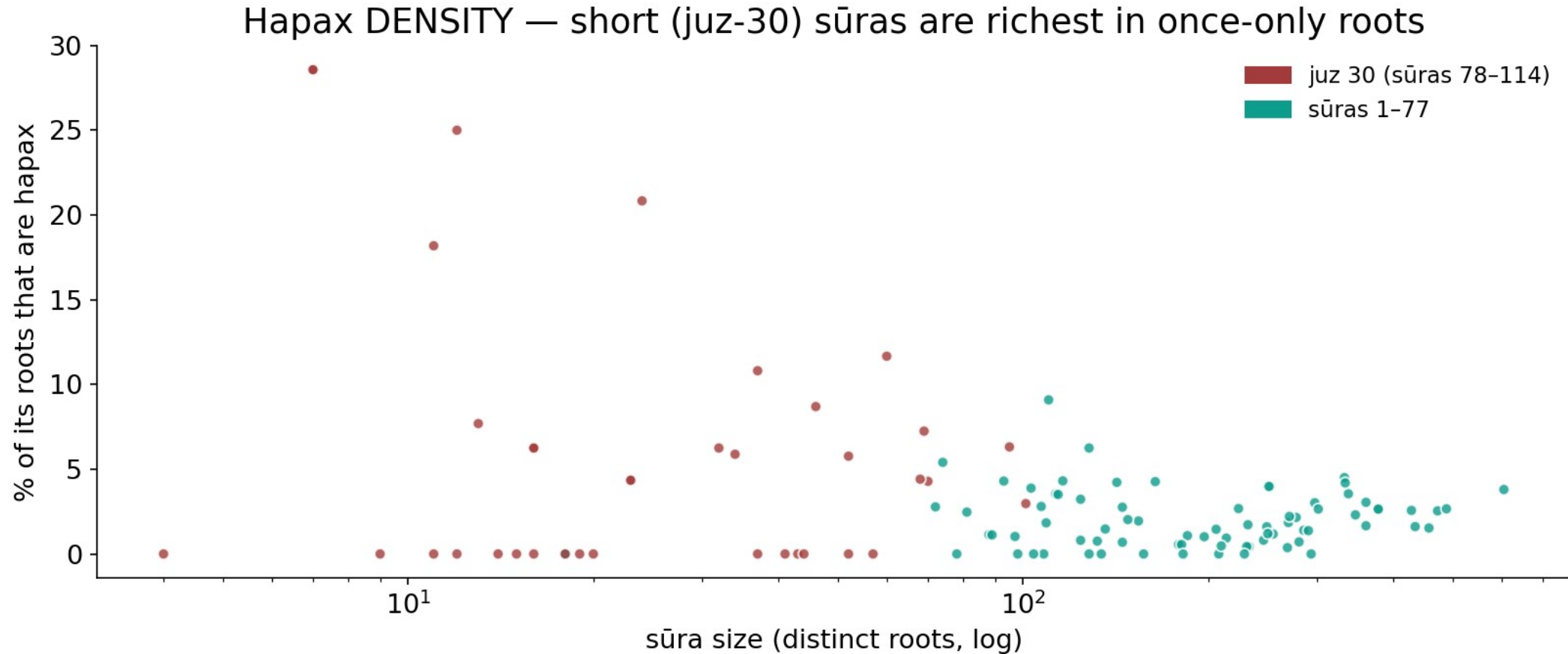
كوثر occurs once, opening al-Kawthar — abundance. Its ROOT ك-ث-ر is common (167 āyahs); only this intensive FORM is unique. (form = 1, root = 167.)

Once-only ROOTS by sūra — where they fall



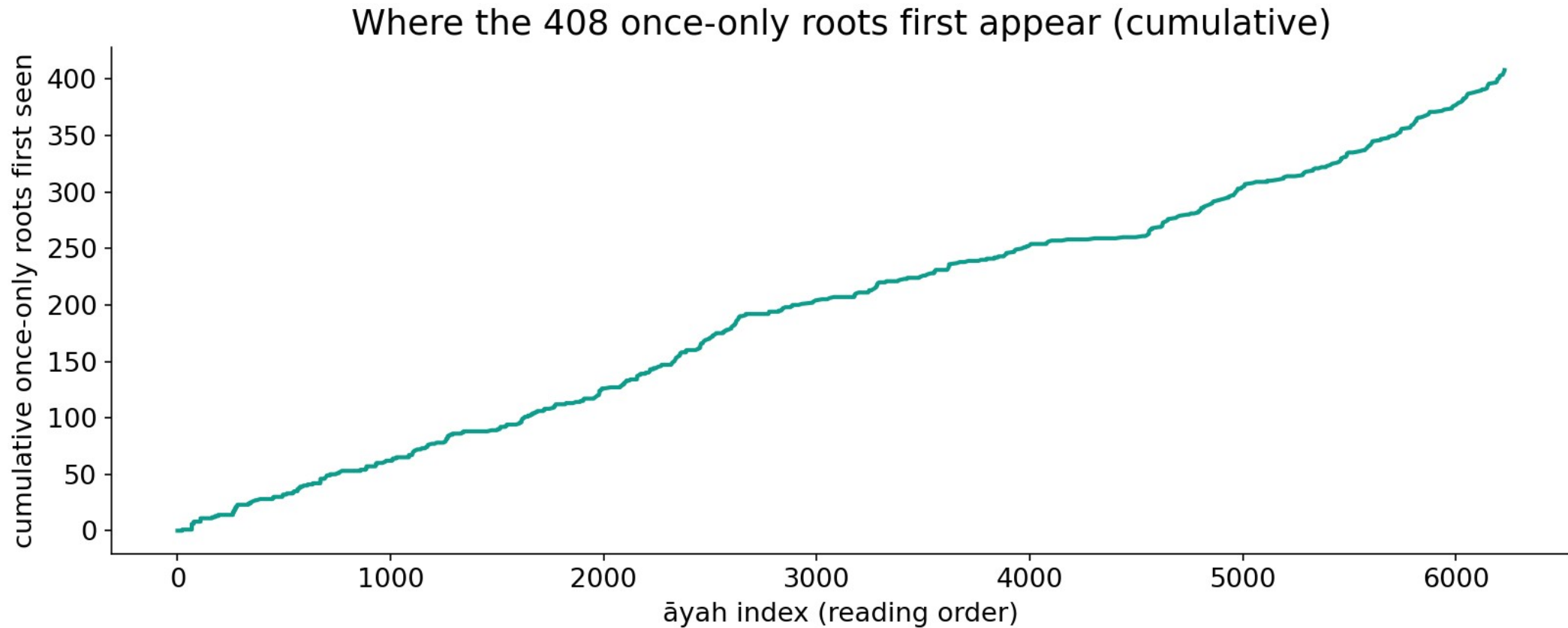
In the data — raw counts are highest in the long sūras (al-Baqara 23, Ṭā-Hā 15, al-Ḥajj 14): length alone puts more unique roots there. Length, not mystery, drives the raw tally.

Density flips it — short sūras are RICHEST



In the data — normalize by sūra size and the juz-30 short sūras dominate. Sūra 108 (al-Kawthar) tops it at ~29% hapax. Brevity concentrates the rare word.

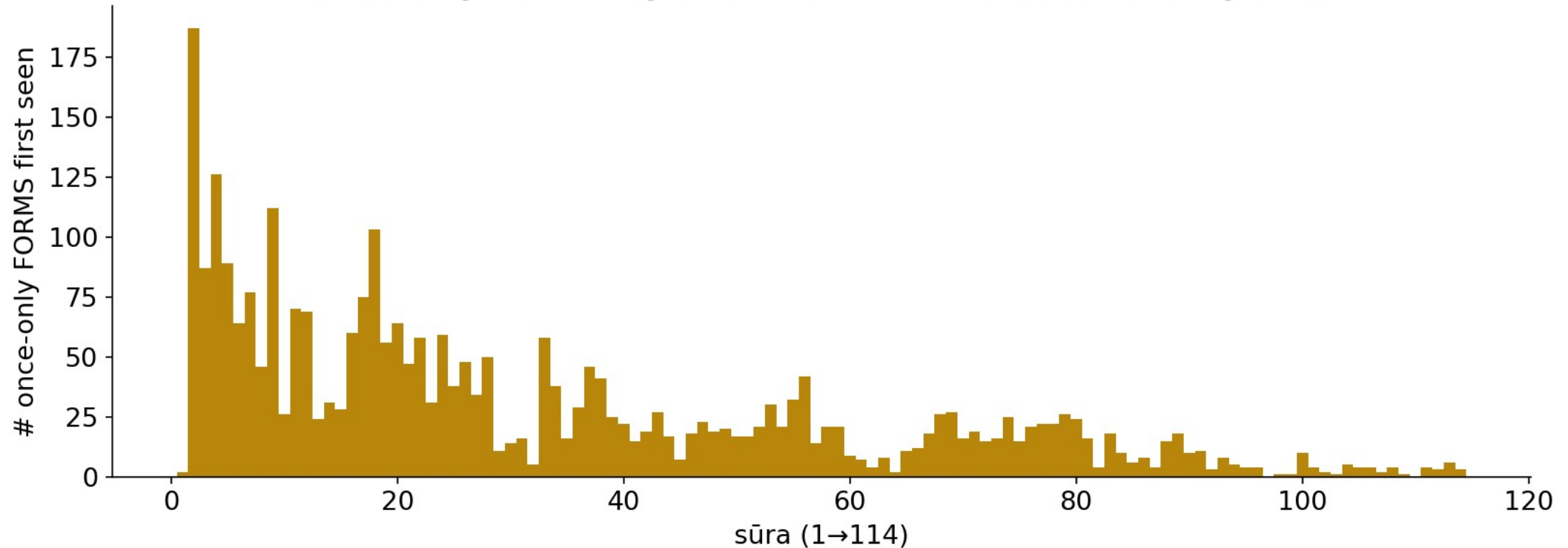
Where the 408 first appear (cumulative)



In the data — new once-only roots accrue steadily through the reading order, with surges in the long early sūras and again in the dense juz-30 tail — the corpus keeps introducing unique vocabulary throughout.

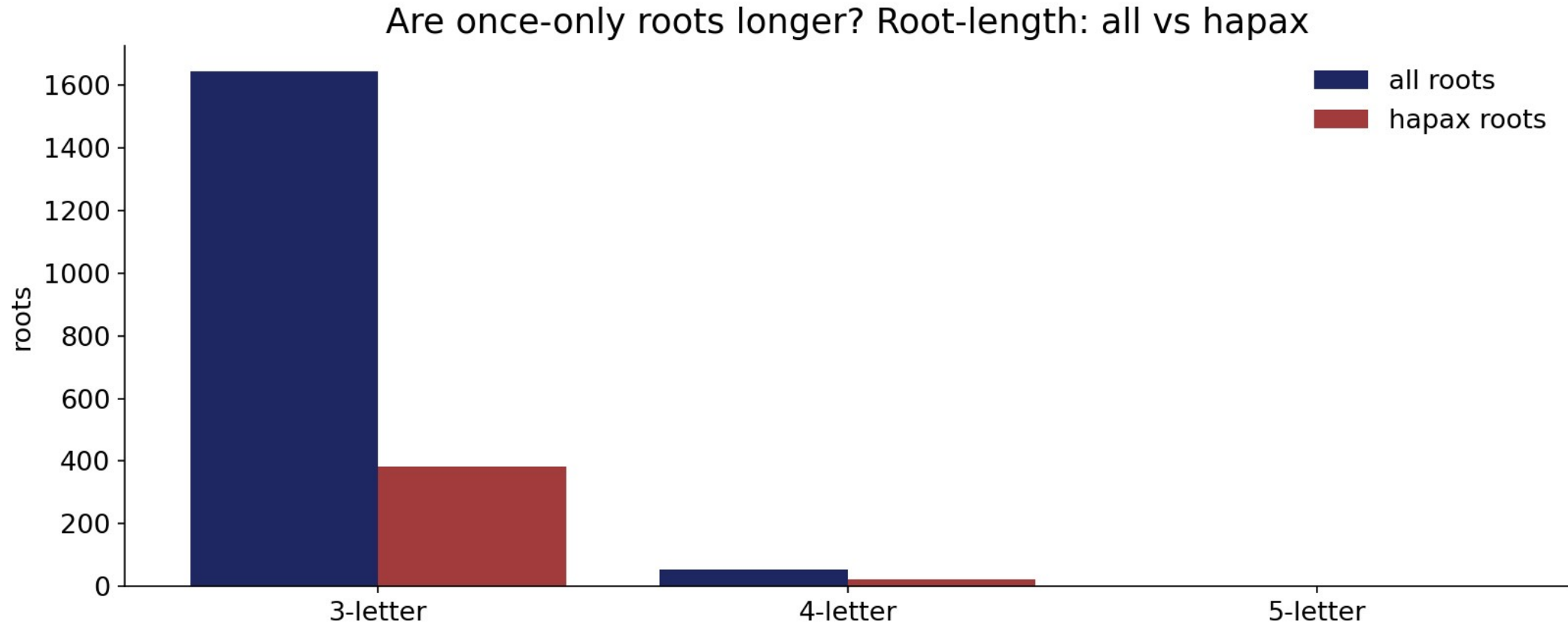
Once-only FORMS by sūra

Once-only FORMS by sūra — 3027 forms occur exactly once



In the data — 3,027 forms occur once; like roots, raw counts track length, but the short late sūras are dense with unique wordings (... عادیات, کنود, هاویه).

Are once-only roots different in shape?



In the data — hapax roots track the overall root-length profile (mostly 3-letter), with a mild lean toward 4-5-letter roots: rarer roots are slightly likelier to be longer/less-templatic.

Gallery — once-only ROOTS in the short sūras (juz 30)

once-only in the whole Qur'an:

صمد — the Eternal / Absolute (112:2)

وعد — the infant girl buried alive (81:8)

عسعس — the night as it recedes (81:17)

كنس — the stars that sweep / recede (81:16)

دهق — a cup brim-full (78:34)

ثجج — water pouring down (78:14)

وهج — a blazing lamp (78:13)

طفف — those who give short measure (83:1)

رين — the rust / stain over hearts (83:14)

عرم — Iram of the pillars (89:7)

Gallery — once-only FORMS (common roots, unique words)

a unique surface form:

كوثر — abundance (root 167× كثر) (108:1)

عاديات — the charging war-steeds (100:1)

كنود — ungrateful to his Lord (100:6)

هاويه — the abyss / Hell (101:9)

مسد — a rope of palm-fibre (111:5)

ضبحا — panting (of the steeds) (100:1)

مبثوث — scattered (moths) (101:4)

منفوش — carded (wool) (101:5)

Once-only ≠ mystery — often it is SPECIFICITY

Specific items

foods, plants, places, materials named once because the topic arises once.

mentioned only here. Rarity = lexical specificity, not hidden code.

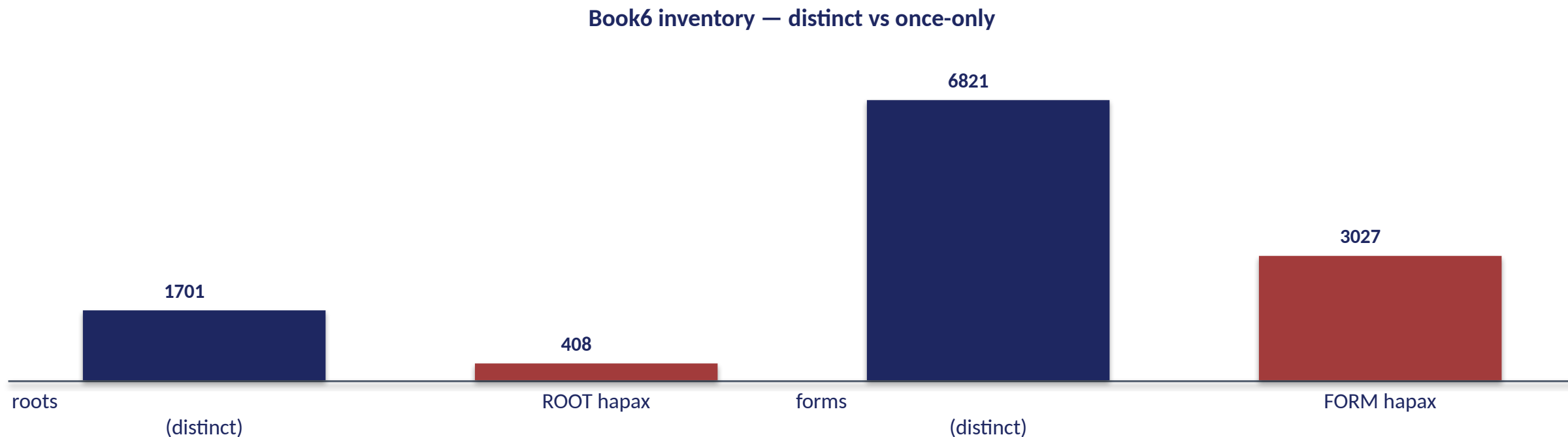
Loanwords / names

proper nouns (Iram) and borrowings naturally appear a single time.

Unique descriptors

a few are unique theological terms — صمد, کوثر — singular by design.

The corpus by rarity — the headline counts

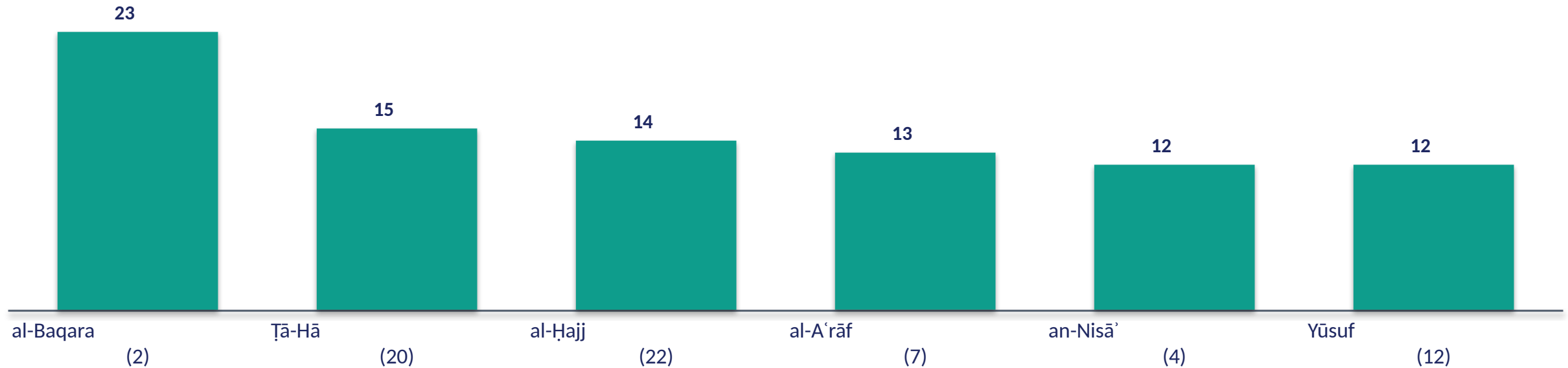


Roughly 1 in 4 roots, and nearly half of forms, are once-only

408 of 1,701 roots (24%) and 3,027 of 6,821 forms (44%) occur exactly once — the corpus is built on a long tail of rare, often single-use vocabulary atop a small core of high-frequency roots.

Which sūras introduce the most once-only roots

Top sūras by # once-only ROOTS first/only seen there



Length leads — but read it against size

The leaders are all long sūras; their raw hapax counts reflect length. The density view (slide 6) is the size-true reading: short sūras are proportionally richer in once-only words.

Resolving the paradox — raw count vs density

RAW COUNT

long sūras hold more hapax — because they hold more of everything

DENSITY

short sūras are richer per word — al-Kawthar ~29%

Two true statements, one size-true

Both charts are real. The raw tally tracks length (a confound); the density (hapax ÷ sūra roots) is the size-true measure and reverses the ranking. The honest claim about 'where rare words live' is the density one — the same normalization discipline used across the whole course.

What the 408 once-only roots actually are

Concrete specifics

foods, animals, materials, body-parts, place-names — most hapax. Named once because mentioned once.

Narrative / proper

one-off names and story-specific terms (Iram, specific peoples, unique events).

Unique descriptors

a small set of singular theological/eschatological terms — aş-Şamad, al-Kawthar, fuṭūr — unique by design.

Reading hapax honestly — the audit

✓ Counts are exact

408 roots / 3,027 forms, reproducible from Book6.

✓ Each is addressable

every hapax carries a sūra:āyah — look it up.

✓ Density is size-true

short-sūra richness holds after normalization.

~ Meaning varies

most hapax are specifics, a few are unique terms.

✗ Not a cipher

a count of one is not a hidden code or miracle.

✗ No over-reading

we report count + address, nothing more.

How this was computed (reproducible)



Full tables provided

Every hapax root and form, with its address, is exported to **Hapax_roots_full.csv** (408) and **Hapax_forms_full.csv** (3,027). Roots are normalized (Persian/Arabic letter variants folded) so the count is stable; the whole pipeline re-runs from Book6.

Takeaway

The once-only word

The Qur'an names 408 roots and over 3,000 forms a single time. Some are unique words for unique things — aṣ-Ṣamad for God, al-Kawthar for the gift; far more are simply the precise word a specific moment required. Rarity is the corpus's norm, concentrated densest in the short sūras.

Read honestly: a count of one, with its address — comprehensive tables attached, nothing over-read.

Quick reference

The numbers

408 once-only ROOTS · 3,027 once-only FORMS · ~24% of roots, ~44% of forms · densest sūra 108 (~29%) · flagship صمد (112:2 root), كوثر (108:1 form).

Where to find it

Full lists: Hapax_roots_full.csv, Hapax_forms_full.csv. Galleries above prioritize the salient short-sūra and theological cases; every figure recomputes from Book6.